Wesley Kwiecinski

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SUMMARY

HPC data engineer of 3+ years. Built Python/MPI batch pipelines and analysis tooling on DOE supercomputers at Argonne National Laboratory, with collaborators at CERN. Built a 32-node pipeline that processed ~1B tokens of LLM training data, analyzed thousands of I/O profiling logs to trace 50% longer runtimes to small-file I/O, and designed relational storage for edge-collected data. M.S. in Computer Science (4.0 GPA), UIC.

TECHNICAL SKILLS

Data Engineering: Python, SQL, Pandas, Polars, NumPy, Parquet, JSON schema design, MariaDB, SQLite, batch data pipelines
Distributed & HPC: MPI, PBS, Slurm, Lustre, Linux; Aurora, Polaris, Perlmutter, Crux supercomputers
Containers & Environments: Apptainer, Podman, Shifter, Conda, Spack, Lmod
Analytics & ML: scikit-learn, PyTorch, torchtitan, Matplotlib, Plotly, Darshan I/O profiling
Other: C/C++, Fortran, Rust; Git/GitHub, pytest, code review, AI-assisted development (Claude Code, Copilot)

EXPERIENCE

Argonne Leadership Computing Facility (ALCF), Argonne National Laboratory *Summer 2026*
Research Aide

- Built a Python/MPI batch pipeline on 32 nodes (PBS) that ingested, transformed, and clustered ~1B tokens of LLM training documents with MiniBatchKMeans, writing output to a custom JSON schema for downstream training; built Matplotlib cluster visualizations to validate groupings.
- Designed an LLM pre-training data pipeline in torchtitan to test document-ordering strategies; difficulty-ordered data reduced training loss faster than other orderings.

Electronic Visualization Laboratory (EVL), University of Illinois Chicago *Fall 2024 – Summer 2026*
Research Aide & M.S. Thesis Research — C²-The-P² Fellow

- Instrumented containerized ATLAS Athena production workflows with Darshan I/O profiling, resolving container-runtime and environment (LCG views, Conda) issues to collect I/O telemetry on CERN nodes and Perlmutter.
- Analyzed thousands of Darshan logs from runs scaling to 64 nodes; traced 50% longer runtimes to wasteful small-file I/O (M.S. thesis).
- Co-authored a CHEP 2026 talk and paper identifying redundant read operations that peak at 10 threads in ATLAS overlay workflows.

Argonne Leadership Computing Facility (ALCF), Argonne National Laboratory *Summer 2025*
Research Aide

- Extended a user-facing performance dashboard for ATLAS workflows on CERN nodes with Darshan-based I/O monitoring (Python), helping software developers analyze I/O performance.

Mathematics and Computer Science Division (MCS), Argonne National Laboratory *Feb 2023 – Aug 2024*
Post-Baccalaureate Appointee (Software Engineering)

- Developed Python tools to analyze and visualize large-scale Parquet log data from the Aurora supercomputer for HPC developers.
- Co-developed Python code-generation tools that emit Fortran CPU–GPU data-transfer code for the Flash-X multiphysics framework (ORCHA); wrote pytest unit tests, joined code reviews, and presented the tools to users at a Flash-X hackathon.

DDILab, Northern Illinois University *Aug 2021 – Dec 2022*
Undergraduate Research Aide

- Designed and implemented a MariaDB database for storing and querying image data from edge devices; co-developed a web-based dashboard for researchers to explore it, and improved accuracy of PyTorch/TensorFlow crosswalk-detection models.

EDUCATION

M.S. Computer Science — University of Illinois Chicago, 4.0 GPA *Aug 2026*
Thesis: Analysis of ATLAS Derivation I/O on High-Performance Computing Systems
B.S. Computer Science — Northern Illinois University, Magna Cum Laude, Minor in Mathematics *Dec 2022*

PUBLICATIONS

- Yildiz, O., Gartung, P., Wang, R., **Kwiecinski, W.** "Characterizing Event I/O Access Patterns in HEP Workflows Using Darshan," Talk & Paper, CHEP 2026.
- Lee, Y., Weide, K., **Kwiecinski, W.**, O'Neal, J., Rudi, J., Dubey, A. "ORCHA – A Performance Portability System for Post Exascale Systems," arXiv:2507.09337, 2025.